

Yunfei Luo

Ph.D. Student in the Halicioğlu Data Science Institute at the University of California San Diego

☎ (413) 404 2124 @ yul268@ucsd.edu 📍 San Diego, CA
🔗 yunfeiluo.github.io in linkedin.com/in/yunfei-luo 🐙 github.com/yunfeiluo

Research Interests

- General** AI/ML, Digital Health, Signal Processing, Multimodal Learning, Multitask Learning, Foundation Model, Reinforcement Learning
- Specific** Yunfei's research focuses on machine learning in the field of digital health, signal processing, multimodal modeling, and multitask learning. He aims to develop a robust and well-generalized methodology that enhances intelligent monitoring, supports timely interventions, and is deployable on edge devices.
- Applications** Mental Health, Substance Usage, Kidney Dialysis, Digital Health Monitoring

Education

- 09.2023 - present **Ph.D. Student in the Halicioğlu Data Science Institute**
University of California, San Diego, CA
‣ Advisor : 📧 Tauhidur Rahman
‣ Research : Machine Learning, Multimodal Learning, Multitask Learning, Digital Health
- 09.2021 - 05.2023 **Master of Science in Computer Science**
University of Massachusetts Amherst, Amherst, MA
‣ Advisor : Ina Fiterau Brostean | 📧 Information Fusion Lab
‣ Courses Taken : Advanced Machine Learning, Neural Networks, Probabilistic Graphical Models, Reinforcement Learning, Advanced Natural Language Processing, Advanced IoT
- 09.2017 - 05.2021 **Bachelor of Science in Computer Science | Bachelor of Science in Mathematics** (Dual Degrees)
University of Massachusetts Amherst, Amherst, MA
‣ Machine Learning and Computing Specializations

Fellowships, Scholarships, Awards, and Honors

- 2023 - 2024 • Institute Fellowship | Halicioğlu Data Science Institute, UC San Diego
- 2021 - 2023 • Bay State Fellowship | Full Tuition Scholarship, UMass Amherst CICS
- 2017 - 2021 • Outstanding Undergrad Course Assistant Award, UMass Amherst CICS
• Chancellor's Award, UMass Amherst

Teaching Experience

- 09.2021 to 06.2023** **Graduate Teaching Assistant, CICS, UMass Amherst**
‣ CS 311 Introduction to Algorithm in Fall 2021 and Spring 2023
‣ CS 590OP Applied Numerical Optimization, and CS 108 Foundation of Data Science in Fall 2022
‣ CS 220 Programming Methodology in Spring 2022
‣ Held office hours, led weekly discussion sessions, graded course works, answered questions on course forum, attended weekly TA meeting for discussing the progress/improvements of the course
Javascript Algorithms Numerical Optimization Data Science Teaching
- 01.2020 to 06.2021** **Undergraduate Course Assistant, CICS, UMass Amherst**
‣ CS 240 Reasoning under Uncertainty in Spring 2021, CS 311 Introduction to Algorithms in Fall 2020, CS 220 Programming Methodology in Spring 2020
‣ Helped with weekly discussion session, graded course works, and answered questions on Piazza
Javascript Algorithms Probability Teaching

Professional Experience

06.2023 to present	Graduate Student Researcher, AI/ML, MOSAIC Lab, UCSD ➤ Research focus on machine learning in signal processing and digital healthcare. Detailed works are presented in the <i>Publications</i> section Research and Development Machine Learning Digital Healthcare Signal Processing Natural Language Processing
06.2022 to 06.2023	Graduate Research Assistant, DS4CG Intern, Machine Learning Engineering, Center for Data Science, UMass Amherst ➤ Developed platform for Building Segmentation and Damage Assessment based on Satellite Imagery ➤ Integrated the Machine Learning pipeline along with User Interface for Imagery Fetching and Labelling ➤ Conducted Research and Experiments with various Finetuning setting and Machine Learning techniques to support selections of final-deployed hyper-parameters and models ➤ Delivered a presentation of the project to the Red Cross (funder and client), resulting in exceptional satisfaction and securing an additional year of funding for the initiative ➤ Worked as Graduate Research Assistant after summer internship Computer Vision Machine Learning Disaster Response Software Development Full-Stack Development
06.2020 to 06.2021	Undergraduate Research Assistant, Core Summer Intern, Data Engineering, Institute for Applied Life Sciences, UMass Amherst ➤ Built pipelines for fMRI data preprocessing and statistical analysis ➤ Conducted Research and Experiments with Machine Learning models for Stroke Detection : Autoencoders, CNNs, and GANs ➤ Worked as Undergraduate Research Assistant after summer internship Machine Learning fMRI Data Preparation

Publications

2026 Published	Toward Foundation Model for Multivariate Wearable Sensing of Physiological Signals, ACM Computing for Healthcare (ACM HEALTH), Journal Yunfei Luo , Yuliang Chen, Asif Salekin, Tauhidur Rahman ➤ Developed NormWear, a foundational model for wearable sensing physiological signals, achieving state-of-the-art performance across 18 healthcare applications spanning 11 public datasets by leveraging generalizable and informative representations. ➤ Proposed a novel representation-alignment framework that aligns physiological and text embeddings, enabling zero-shot inference for previously unseen health applications in wearable sensing. ➤ Conducted comprehensive nonlinear dynamic analysis of intermediate model representations, offering quantifiable insights into internal processes and fostering trust in healthcare-related inferences. Signal Processing Foundation Model Natural Language Processing Machine Learning Neural Networks Digital Health
2026 Published	Personalized entropy-informed deep learning for identifying opioid misuse, Nature Mental Health, Journal Yunfei Luo , Iman Deznabi, Bhanu Gullapalli, Mark Tuomenoksa, Eric Garland, Madalina Fiterau, Tauhidur Rahman ➤ Leveraging transformer based time series model and dynamic branching to predict the stress, pain, and craving of Opioid users based on heart beat data collected from wearable sensor. ➤ Leveraging topological features from trajectory of inferential affective states across days and natural language processing technique to predict the risk of having opioid misuse behavior. Signal Processing Natural Language Processing Heart Rate Variability Affective States Opioid Clinical Treatment Machine Learning Multimodal Fusion Digital Health
2026 Published	Automated Dysphagia Screening Using Noninvasive Neck Acoustic Sensing, IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), Conference Jade Chng , Rong Xing (Co-1st Author), Yunfei Luo (Co-1st Author) , Kristen Linnemeyer-Risser, Tauhidur Rahman, Andrew Yousef, Philip A Weissbrod ➤ We present a noninvasive, acoustic-based machine learning framework that detects dysphagia with high accuracy, offering a scalable tool for monitoring pharyngeal health. Signal Processing Acoustic Signal Data Collection Clinical Treatment Applied Machine Learning Digital Health

2025 Published	Sounding out Dysphagia : Using a Digital Stethoscope to Assess Audiometric Differences in Swallowing, The Laryngoscope, Journal Andrew Yousef, Sacha Moufarrej, David Zeng, Bao Luu, Rong Xing, James S. Crook, Yunfei Luo, Kristen Linnemeyer-Risser, Tauhidur Rahman, Philip A Weissbrod ➤ We demonstrated that digital stethoscope-based audiometric analysis can objectively distinguish normal and dysfunctional swallows during FEES. ➤ We identified key acoustic biomarkers, including duration, amplitude, and frequency, that correlate with swallow dysfunction severity, enhancing clinical evaluation accuracy. Signal Processing Digital Health Dysphagia Acoustics Non-invasive
2025 Published	Opioid Misuse Detection from Cognitive and Physiological Data with Temporal Fusion Deep Learning, Drug and Alcohol Dependence, Journal Bhanu Gullapalli, Yunfei Luo, Tauhidur Rahman, Eric L. Garland ➤ We used deep learning to detect opioid misuse from cognitive and physiological data. ➤ Behavioral responses were stronger predictors of misuse than physiological signals. ➤ Our model accurately detected opioid misuse. Signal Processing Machine Learning Digital Health Addiction Psychophysiology Chronic Pain
2024 Published	Dynamic Clustering via Branched Deep Learning Enhances Personalization of Stress Prediction from Mobile Sensor Data, Nature Scientific Report, Journal Yunfei Luo, Iman Deznabi, Abhinav Shaw, Tauhidur Rahman, Madalina Fiterau Brostean ➤ Proposed a novel approach to modeling the Multimodal data, got results that outperform SOTA ➤ Modeled personalized and populational characteristics with Multitask Learning and Dynamic Clustering Method respectively ➤ Evaluated the performance under cold-start setting, to support the robustness of our method Multi-modal Fusion Methods Mental Health Machine Learning Neural Networks Time Series
2024 Published	Start Simple : Progressive Difficulty Multitask Learning, The Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL) 2024 SRW Proceedings, Conference Yunfei Luo, Yuyang Liu, Rukai Cai, Tauhidur Rahman ➤ Introducing sub-tasks with progressive difficulties to improve the performance on main task. Inspired by how people learn. Experimented on text classification, sentiment analysis, and argument mining Natural Language Processing Multi-task Learning Machine Learning Neural Networks
2023 Published	Agent Performing Autonomous Stock Trading under Good and Bad Situations, AI for Agent-Based Modelling (AI4ABM) in International Conference on Learning Representations (ICLR) 2023, Workshop Yunfei Luo, Zhangqi Duan ➤ Researched and experimented with different reinforcement learning algorithms to train agents to perform stock trading. The methods are evaluated under scenarios with stable and non-stable market Reinforcement Learning Machine Learning Neural Networks

Papers Under Preparation

2026 Under Review	Toward World Modeling of Physiological Signals with Chaos-Theoretic Balancing and Latent Dynamics, -, Conference Yunfei Luo, Xi Chen, Yuliang Chen, Lanshuang Zhang, Md Mofijul Islam, Siwei Zhao, Peter Kotanko, Subhasis Dasgupta, Andrew Campbell, Rakesh Malhotra, Tauhidur Rahman ➤ We introduce a framework for quantifying and balancing the dynamical diversity of pretraining signals datasets, improving the robustness and transferability of learned representations. ➤ We propose NormWear 2, which combines prior dynamics and energy-based adaptation in latent space to enable multi-step forecasting and generative modeling of physiological signals. ➤ We develop a multidimensional assessment across time, frequency, and latent representation spaces, demonstrating effectiveness on diverse physiological datasets at multiple time scales. Signal Processing Foundation Model World Model Machine Learning Neural Networks Digital Health
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2026	Counterfactual Simulation–Driven Reinforcement Learning for Acute Intervention in Intra-dialytic Hypotension, -, Journal
Preparing	Yunfei Luo, Siwei Zhao, Subhasis Dasgupta, Joseph M Mahaffy, Peter Kotanko, Rakesh Malhotra, Tauhidur Rahman ➢ Investigated counterfactual exploration across value-based, policy-based, and actor–critic RL methods, analyzing off-policy learning effects and resulting intervention behaviors. ➢ Evaluated RL agents in simulated dialysis sessions, showing more conservative fluid removal strategies with reduced intradialytic hypotension (IDH) incidence while preserving fluid removal performance without manual trade-off tuning. ➢ Simulated collaborative clinician–agent decision-making, demonstrating reduced IDH incidence with clinically appropriate fluid removal outcomes. Signal Processing Reinforcement Learning Nephrology Clinical Treatment Machine Learning Digital Health
2026	Real Time Intradialytic Hypotension Forecasting Leveraging Modern Deep Learning Methods with Multimodal Fusion of Physiological Data and Electronic Health Records, npj Digital Medicine, Journal
Await Decision	Yunfei Luo, Siwei Zhao, Subhasis Dasgupta, Joseph M Mahaffy, Peter Kotanko, Tauhidur Rahman (Equal Senior Author), Rakesh Malhotra (Equal Senior Author) ➢ Novel multimodal modeling on physiological data, dialysis machine parameter, and electronic health records. Aiming to accurately forecasting hypotension 1 hour ahead in dialysis patient, and to provide comprehensive insights to nurse for intervention recommendation. Signal Processing Nephrology Clinical Treatment Machine Learning Multimodal Fusion Digital Health

Patents

- 2025 US2025/016369 - System and Method for Personalized Closed-Loop Opioid Addiction Management with Mobile and Wearable Sensing of Administrations, Affective States, and Misuse Risk Scores (**Full US Utility patent application filed on Feb 18, 2025**)

Talks

- 2026 • “Toward Foundation Model for Multivariate Wearable Sensing of Physiological Signals”, Lightning Talk at **Conference of Naval Applications of Machine Learning (NAML)**, March 3rd
- 2025 • “Foundation Model for Multivariate Wearable Sensing of Physiological Signals”, **Google Research Journal Club**, May 27th
- “Health AI, foundation models, agents, and related research”, **Dexcom Journal Club**, April 18th
- 2023 • “Integrating Machine Learning Models to Your Applications”, **Workshop at SackHacks V**, November

Professional Services

- Reviewer**
- Neural Information Processing Systems (NeurIPS), 2026
 - Main Track 2026
 - International Conference on Learning Representations (ICLR), 2026
 - Main Track 2026
 - Workshop on Time Series in the Age of Large Models, International Conference on Learning Representations (TSALM), 2026
 - Workshop on AI with Recursive Self-Improvement (RSI), 2026
 - International Conference on Machine Learning (ICML), 2025 to 2026
 - Main Track 2026
 - Workshop on Foundation Models for Structured Data, International Conference on Machine Learning (FMSD), 2025 to 2026
 - Conference on Computational Natural Language Learning (CoNLL), 2024 to 2026
 - Proceedings of ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT), 2024 to 2026

Professional Certifications

2022 IBM Artificial Intelligence Engineering Professional Certificate
2022 Neuroscience and Neuroimaging Specialization Certificate
2020-2023 TensorFlow Developer Professional Certificate

References

Tauhidur Rahman

Assistant Professor of Data Science
University of California San Diego
Director of MOSAIC Lab
@ trahman@ucsd.edu

Benjamin Smarr

Associate Professor of Data Science
University of California San Diego
Department of Bioengineering
@ bsmarr@ucsd.edu

Tom Bernardin

Data Scientist
University of Massachusetts Amherst
Director of the Center for Data Science
@ tbernard@cs.umass.edu

Rakesh Malhotra

Assistant Professor of Medicine
University of California San Diego
Division of Nephrology-Hypertension
@ r3malhotra@health.ucsd.edu

Madalina Fiterau Brostean

Assistant Professor of Computer Science
University of Massachusetts Amherst
Lead of the Information Fusion Lab
@ mfiterau@cs.umass.edu

Daniel Sheldon

Associate Professor of Computer Science
University of Massachusetts Amherst
College of Information and Computer Science
@ sheldon@cs.umass.edu

Eric L. Garland

Professor in the Department of Psychiatry
University of California San Diego
Director of UCSD ONEMIND
@ egarland@health.ucsd.edu

Subhansu Maji

Associate Professor of Computer Science
University of Massachusetts Amherst
Co-Director of Computer Vision Lab
@ smaji@cs.umass.edu